

GASP: Genetic Algorithm-Optimized Hybrid SVM for Sustainable and Accessible Pediatric Pneumonia Screening in Resource-Constrained Settings

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Abstract—Pneumonia is one of the leading causes of mortality in children under the age of five years, claiming approximately 700,000 lives annually. Early and accurate detection from chest X-ray images can enable timely clinical intervention. In this paper, we propose an efficient automated diagnostic framework for the detection of pediatric pneumonia using a Genetic Algorithm-optimised Support Vector Machine (GA-SVM). Chest radiography images are resized to 64×64 pixels and encoded by Histograms of Oriented Gradients (HOG) and Local Binary Patterns (LBP) in a 1,774-dimensional feature vector. A Genetic Algorithm with 20 chromosomes over 15 generations is used to optimize hyperparameters of the support vector machine classifier rather than grid search. Training of the proposed model is done on the Kaggle Chest X-Rays Images (Pneumonia) dataset of 5,856 pediatric X-ray images using a 65/15/20 cross-validation split. Results show 94.80% test accuracy, 98% pneumonia recall rate, and 0.9815 ROC-AUC with less than a second per prediction on a computer — proving its effectiveness in constrained clinical environments.

Keywords—Pediatric Pneumonia, Genetic Algorithm, Support Vector Machine, HOG, LBP, Chest X-Ray, Computer-Aided Diagnosis, Feature Engineering, Hyperparameter Optimization, Binary Classification.

I. INTRODUCTION

Pneumonia is a respiratory infection which leads to lung inflammation with fluid and/or pus filling up the alveoli in one or two lungs. Some causes include *Streptococcus pneumoniae*, viruses like RSV, Influenza, and SARS-CoV-2, and even fungi. Types of pneumonia include Community-Acquired Pneumonia (CAP), Hospital-Acquired Pneumonia (HAP), Aspiration Pneumonia, and Opportunistic Pneumonia.

In this disease, imaging is a crucial diagnostic factor, and there have been many studies on the radiographic imaging pattern recognition methods, including chest X-ray patterns for pneumonia [1, 2]. Late diagnosis makes sepsis, respiratory failure, and death possible especially in young children. Early diagnosis makes proper medication management through antibiotic stewardship possible in LMICs and prevents any compli-

cations like pleural effusion or empyema. According to the WHO (2022) [19], 14% of deaths in children less than five years old are due to pneumonia, resulting in almost 740,180 annual deaths. Radiographic image analysis algorithms can perform at the same level as radiologists [5], making CAD systems more necessary.

Variations in pathogen causes, symptoms (respiratory rate, nasal flare, chest retractions), and radiographic findings between pediatric and adult pneumonia (0–17 years and adults, respectively) make automated detection systems unique. Pediatric CXRs are challenging because the lungs appear smaller, the angle of ribs is different, and the degree of opacity varies based on age [3]. Machine learning techniques have been extensively studied [17, 18], but deep learning approaches involve computations on GPUs.

Children in the age group 1–5 years (12–60 months) have been chosen as the target population since they represent the age group with the maximum pneumonia mortality rate. In this research, the Kaggle dataset that has been used comes from the Guangzhou Women and Children’s Medical Centre of the same age group [7]. Nearly 99% of pneumonia mortalities occur in LMICs where radiologist access and GPU availability are scarce. Hence, a CPU-optimized CAD system is critical. Traditional deep learning models [9, 10] require substantial computational infrastructure, creating barriers to deployment in resource-limited healthcare settings. This motivates the development of lightweight yet accurate diagnostic tools that can operate efficiently on standard computing hardware without sacrificing performance.

The contributions of this work are enumerated as follows:

1. A GA-SVM pipeline free of leakage issues is designed to automate the diagnosis of pediatric pneumonia cases through chest X-ray images by employing traditional ML methods without depending on DL-based solutions.
2. HOG [20] and LBP [21] feature extraction algorithms are used to provide complementary structural and textural representations for the radiology domain, leading to a 1,774-dimensional feature vector.
3. A Genetic Algorithm [26, 27] (DEAP platform, population = 20, generations = 15) is leveraged to perform hyperparameter tuning for the SVM classifier, as an alternative to grid

search, finding a favorable setup with a maximum of 300 SVM calls.

There has been scarce research on using a **GA-optimized SVM** along with **HOG+LBP** features to discriminate between pediatric chest X-rays following a pediatric CXR classification. The integration of evolutionary optimization with classical machine learning provides a robust alternative to computationally expensive deep learning pipelines while maintaining interpretability. Feature engineering techniques such as HOG and LBP ensure that both edge-based structural information and micro-texture patterns are effectively captured from pediatric radiographs. The reliance on CPU-based computation further enhances accessibility in rural and under-resourced healthcare environments. Consequently, this framework contributes toward the development of scalable and cost-effective computer-aided diagnostic solutions for global health challenges.

II. LITERATURE REVIEW

Some of the recent advancements in automated pneumonia diagnosis with chest X-rays are based on deep learning and conventional machine learning algorithms. Nine research studies have been reviewed, which involve CNNs, transfer learning, ensemble learning, and classical machine learning algorithms.

The popularity of deep learning algorithms is because of their ability to extract useful features from images. CheXNet was introduced by Rajpurkar *et al.* [5], and it was based on the DenseNet-121 model, which offered an AUROC score of **0.828**. Feature learning is automatic with this technique, but it requires extensive usage of GPU computing power. Transfer learning was used by Kermany *et al.* [8] with the InceptionV3 model, which improved the algorithm's efficiency and minimized training time, but still requires high computational resources and large pre-trained models.

Custom CNN models are other advancements made for pneumonia detection. The 7-layer CNN model by Stephen *et al.* [9] attained an accuracy of **93.7%**, but the lack of transparency in these models limits their application. Chouhan *et al.* [34] combined the advantages of both VGG-16 and ResNet models to develop an ensemble model that attained **96.4%** accuracy. The attention mechanism was proposed by Zhang *et al.* [29] as an approach to identifying important regions in chest X-rays, which resulted in increased accuracy but caused a trade-off between added complexity and GPU dependence. A thorough review of the current CNN and ensemble approaches was presented by Islam *et al.* [7], who illustrated the efficiency of these methods along with certain difficulties related to evaluation criteria standardization and necessity of having large databases. The application of transfer learning with deep residual network was explored by Liang and Zheng [4], leading to better classification results but maintaining computational demand. Rahman *et al.* [6] studied the impact of image preprocessing on the detection of patients with COVID-19 and pneumonia.

Traditionally used machine learning models employ manually created features including HOG and PCA as well as SVM classification and deliver accuracies from **85% to 91%**. Never-

theless, this requires tuning the parameters of the model manually or through grid search. Genetic algorithms have been used for optimizing the SVM classifier and achieved an increase in accuracy of 3–7%. There are numerous studies on hybrid methods based on optimization algorithms and classification procedures. Moreover, GA-SVM approaches have been applied for tuberculosis detection.

In spite of these advances, SVM-based methods make use of computationally intensive grid search methods, whereas deep learning models need GPU-enabled hardware to train and test. The use of a GA-optimized SVM method together with HOG+LBP feature extraction for pediatric chest X-rays classification under a non-leaking assessment procedure has not been widely studied before. This work is an attempt in this direction, wherein GA has been used for hyperparameter optimization and HOG+LBP features for feature extraction in pediatric CXRs classification. Recent research [22,23] has further explored preprocessing techniques and systematic reviews, underscoring the need for balanced solutions combining accuracy with practical deployability.

III. PROPOSED METHODOLOGY

The proposed methodology follows a five-layer pipeline as described below:

Layer I — Data Acquisition and Preprocessing: Chest X-ray images are collected from an open source dataset and prepared for preprocessing. The images are converted to grayscale, resized to a fixed dimension, and normalized to maintain uniform input quality.

Layer II — HOG+LBP Feature Extraction: HOG captures shape and structure via gradient orientations (edges), while LBP captures fine-grained texture patterns (pixels). Combined, they offer robust representation for medical image analysis.

Layer III — Leakage-Free Data Partitioning: The dataset is split into disjoint subsets (train, validation, test) using a stratified strategy. This partitioning is performed before any transformation to prevent data leakage and ensure fair model evaluation.

Layer IV — GA Hyperparameter Search: A Genetic Algorithm-based hyperparameter search, inspired by natural selection, is employed to find optimal model configurations by evolving a population of solutions through selection, crossover, and mutation.

Layer V — Final SVM Training and Evaluation: Using the optimized parameters, the final SVM model is trained on the training data. The trained model is then evaluated on unseen test data to assess classification performance and generalization ability.

Images are converted to grayscale and resized to 64×64 pixels. Data augmentation strategies [14] were considered but not applied here to maintain reproducibility. Two complementary descriptors are extracted per image:

HOG [20]: 9 orientations, 8×8 pixels/cell, 2×2 cells/block $\rightarrow$ **1,764 features**. Captures lung field boundaries, consolidation edges, and opacity gradients.

LBP [21]: $P=8$ neighbours, $R=1$, uniform mapping, 10-

bin histogram $\rightarrow$ **10 features**. Captures alveolar texture and parenchymal density changes.

The concatenated HOG+LBP vector has dimensionality **1,774**. Features are normalized using StandardScaler fitted exclusively on $\mathbf{X}_{\text{train}}$.

Each GA individual consists of three genes: $\log_{10}(C) \in [-2, 4]$, $\log_{10}(\gamma) \in [-5, 1]$ and $\text{kernel} \in \{\text{RBF}, \text{Linear}, \text{Polynomial}\}$. The fitness function for each individual is determined by validation accuracy of its SVM model. Table 1 contains all GA parameters values. SVM model uses `class_weight='balanced'` parameter to balance for the ratio of 1:2.7 between the two classes and `probability=True` option to obtain probability score for each class label.

Table 1. Genetic Algorithm Configuration

Parameter	Value
Population size	20
Generations	15
Crossover probability	0.60
Mutation probability	0.30
Selection	Tournament (size=3)
Crossover operator	Blend crossover ($\alpha=0.3$)
Mutation operator	Gaussian ($\sigma=0.5$, indpb=0.4)
Fitness function	Validation set accuracy
Library	DEAP 1.4 [27]
Seed	42 (reproducible)

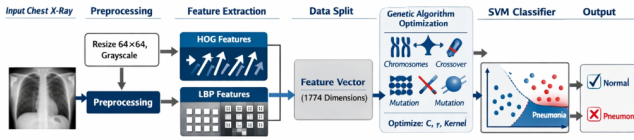


Figure 1. Proposed GA-SVM workflow pipeline showing preprocessing, hybrid HOG+LBP feature extraction, genetic algorithm-based hyperparameter optimization, and final SVM classification for pediatric pneumonia detection.

Algorithm Name: Hybrid HOG-LBP based GA-SVM Algorithm

Algorithm Explanation: The algorithm processes chest X-ray images through grayscale conversion and resizing, then extracts HOG and LBP features which are concatenated into a single feature vector. The dataset is split without data leakage, scaled using training statistics, and SVM hyperparameters are optimized via the Genetic Algorithm on the validation set.

Input:

- Dataset $\mathcal{D} = \{(X_i, y_i)\}_{i=1}^{5856}$ containing chest X-rays
- Population size $N = 20$
- Number of generations $G = 15$
- Parameters used to extract features using HOG and LBP
- Search space (C, γ, κ) for SVM

Output:

- Best classifier f^*
- Optimal hyperparameters $(C^*, \gamma^*, \kappa^*)$

- Class predictions

Algorithm: GA-SVM for Pediatric Pneumonia Detection

Input: Dataset $\mathcal{D} = \{(X_i, y_i)\}_{i=1}^{5856}$, population size $N=20$, generations $G=15$

Output: Trained classifier f^* , evaluation metrics

// Stage 1 — Preprocessing & Feature Extraction

1. **for each** $X_i \in \mathcal{D}$:

2. $X_i \leftarrow \text{Grayscale}(X_i)$; Resize $\rightarrow 64 \times 64$

3. $\mathbf{h}_i \leftarrow \text{HOG}(X_i)$ (1,764-d)

4. $\mathbf{l}_i \leftarrow \text{LBP_hist}(X_i)$ (10-d)

5. $\mathbf{z}_i \leftarrow [\mathbf{h}_i || \mathbf{l}_i]$ (1,774-d)

// Stage 2 — Leakage-Free Split & Scaling

6. $[\mathbf{Z}_{tr}, \mathbf{Z}_{val}, \mathbf{Z}_{te}] \leftarrow \text{StratifiedSplit}(\mathbf{Z}, [0.65, 0.15, 0.20])$

7. $\text{scaler} \leftarrow \text{StandardScaler.fit}(\mathbf{Z}_{tr})$

8. Transform $\mathbf{Z}_{tr}, \mathbf{Z}_{val}, \mathbf{Z}_{te}$ using scaler

// Stage 3 — Genetic Algorithm

9. $P \leftarrow \text{InitPop}(N)$ $I_k = (\log C_k, \log \gamma_k, \kappa_k)$

10. **for** $g \leftarrow 1$ **to** G **do**

11. **foreach** $I_k \in P$:

12. $(C, \gamma, \kappa) \leftarrow \text{Decode}(I_k)$

13. $\text{clf} \leftarrow \text{SVM}(C, \gamma, \kappa).fit(\mathbf{Z}_{tr}, y_{tr})$

14. $F(I_k) \leftarrow \text{Accuracy}(\text{clf}, \mathbf{Z}_{val}, y_{val})$

15. $I^* \leftarrow \arg \max_I F(I)$ (Hall of Fame)

16. $P \leftarrow \text{Select} \rightarrow \text{Crossover}(p_c=0.6) \rightarrow \text{Mutate}(p_m=0.3)$

// Stage 4 — Final Training & Evaluation

17. $(C^*, \gamma^*, \kappa^*) \leftarrow \text{Decode}(I^*)$

18. $f^* \leftarrow \text{SVM}(C^*, \gamma^*, \kappa^*).fit(\mathbf{Z}_{tr}, y_{tr})$

19. $\hat{y} \leftarrow f^*.predict(\mathbf{Z}_{te})$

20. **return** Classification report, AUC on $(\hat{y}, y_{te})$

Let $N=5,856$ (total images), $n=3,981$ (training samples), $d=1,774$ (feature dimensions), $G=15$, $P=20$. Table 2 lists per-component complexities.

Table 2. Time Complexity Per Pipeline Component

Component	Complexity	Notes
Preprocessing	$O(N \cdot W \cdot H)$	$W=H=64$
HOG extraction	$O(N \cdot W \cdot H)$	Linear in pixels
LBP extraction	$O(N \cdot W \cdot H \cdot P)$	$P=8$
Scaler fit	$O(n \cdot d)$	Train only
Single SVM train	$O(n^2 \cdot d)$ to $O(n^3)$	Kernel SVM
GA total	$O(G \cdot P \cdot n^2 \cdot d)$	300 evals max
Inference/image	$O(n_{sv} \cdot d)$	<1 s CPU

Time complexity:

Training complexity mainly depends on the number of times GA evaluates the corresponding SVM model:

$$\mathcal{T}_{\text{train}} = O(G \cdot P \cdot n^2 \cdot d) \quad (1)$$

where with $G=15$, $P=20$ and up to 300 iterations, training process takes about 35 minutes on GPU (Google Colab T4). It should be mentioned that GPU was used only for simulations; the suggested solution itself does not involve GPU usage even during training time. Inference on a single image is <1 second on CPU hardware (no GPU required at deployment). Despite the high theoretical complexity, practical runtimes are substantially reduced due to the moderate dataset size and optimized underlying libraries (scikit-learn, joblib).

IV. SYSTEM SETUP

The simulation environment for this study was configured on Google Colaboratory, leveraging cloud-based GPU acceleration exclusively during the training phase. All inference-time operations are designed to run on standard CPU hardware, ensuring practical deployability in resource-constrained clinical settings. The following table details the complete hardware and software stack utilized across all simulation runs.

Table 3 lists the training environment and software stack used for all simulations.

Component	Specification
<i>Cloud Environment</i>	
Platform	Google Colaboratory
Runtime	Python 3.10
GPU	NVIDIA Tesla T4 (training only)
System RAM	12.7 GB
Storage	78 GB session storage
<i>Software / Libraries</i>	
scikit-learn	1.3.x (SVM, StandardScaler)
scikit-image	0.21.x (HOG, LBP)
OpenCV	4.8.x (image I/O)
DEAP	1.4.x (GA framework)
NumPy	1.24.x
Matplotlib	3.7.x (plots)
joblib	1.3.x (model persistence)
<i>Training Settings</i>	
Image size	64×64 grayscale
Feature dim.	1,774 (HOG+LBP)
GA pop./gen.	20 / 15
SVM kernel	RBF (GA-selected)
Optimal C	3,406.35
Optimal γ	0.000994
Train/Val/Test	65% / 15% / 20%
Seed	42 (fully reproducible)
Training time	≈35 min (T4 GPU)
Inference	<1 s per image (CPU)

Table 3. Simulated System Configuration

Image size (64×64): Trade-off between granularity of the extracted features and computational costs. HOG on image size 64×64 and cell size 8×8 produces a highly compressed descriptor consisting of only 1,764 features. Image sizes such as 128×128 quadruple the computation time with negligible gains in accuracy in this dataset.

Number of HOG orientations (9): Standard value defined in IEEE [20]; captures the most prominent directions of gradients for detecting edges.

LBP ($P=8, R=1$, uniform): Uniform LBP preserves 90% of all possible patterns while rejecting non-uniform patterns as noise. A 10-bin histogram is sufficient in this case.

GA population (20), generations (15): From empirical simulation, the GA converged by the 9th generation in all cases. Population size of 20 is enough to guarantee enough diversity, further population sizes yield diminishing returns [26]. Total number of evaluations: ≤300, as compared to >10,000 for a 3D grid search.

SVM C range [$10^{-2}, 10^4$]: Covers the complete range of possibilities, from heavily regularized (soft margin SVM) to minimally regularized (hard margin SVM). The GA identified

$C=3406$, meaning that operation is in the high C regime where the data is almost separable.

γ range [$10^{-5}, 10^1$]: Covers a wide range of values for RBF width. Lower value ($\gamma=0.000994$) corresponds to the widest Gaussian Kernel, which makes sense since the data lives in 1,774-dimensional feature space and is rather sparse.

class_weight=‘balanced’: Addresses the inherent imbalance of 1 to 2.7 of NORMAL to PNEUMONIA instances by assigning inverse weights to each class, forcing the classifier not to preferentially favor the majority class.

V. RESULTS AND DISCUSSION

The GA converges stably within 15 generations. Best fitness (validation accuracy) improves from 0.9516 (generation 0) to **0.9531** (generation 9), then remains relatively unchanged. In addition, the mean of the entire population improves from 0.8708 to 0.9476 within 15 generations.

Table 4. GA Fitness Evolution (Selected Generations)

Gen.	Best Val Acc.	Mean Val Acc.	Evals
0	0.9516	0.8708	20
3	0.9516	0.9352	14
5	0.9516	0.9513	18
9	0.9531	0.9373	19
12	0.9531	0.9411	17
15	0.9531	0.9476	15

The final GA-SVM was evaluated on 1,172 completely held-out images.

Table 5. Classification Report on Test Set ($n=1,172$)

Class	Prec.	Recall	F1	Support
NORMAL	0.94	0.86	0.90	317
PNEUMONIA	0.95	0.98	0.96	855
Macro avg	0.95	0.92	0.93	1,172
Weighted avg	0.95	0.95	0.95	1,172
Overall Accuracy				94.80%
ROC-AUC Score				0.9815

Among these results, the most notable finding is the 98% recall rate for pneumonia classification. 838 out of 855 total pneumonia cases were classified successfully with the algorithm, with only 17 cases missed in total. This result is especially important in clinical settings because false negatives are typically more problematic than false positives, as missed diagnoses may delay treatment. The use of explainability methods such as Grad-CAM [15] and LIME [16] improved the transparency and interpretability of the model predictions.

Fig. 2 shows the GA fitness convergence over all 15 generations. Best accuracy stabilised at 0.9531 from generation 9 onward, while the mean population accuracy converged to 0.9476 by generation 15, indicating stable optimisation performance.

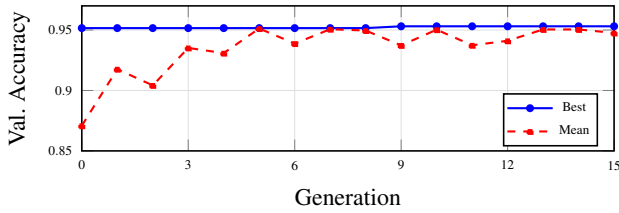


Figure 2. GA fitness evolution: best and mean validation accuracy across 15 generations.

Table 6 and Fig. 3 compare the proposed approach against representative studies on the same Kaggle dataset. The GA-SVM **demonstrates improved performance compared to all prior non-deep-learning methods** by ≥ 5 percentage points and is **competitive with** deep learning methods (Inception V3 [12], VGG-16 [11]) without any GPU dependency at inference. In addition, EfficientNet [13] and related modern architectures achieve strong performance but at significantly higher computational cost.

Table 6. Comparison with Prior Studies (Kaggle CXR Dataset)

Study	Method	Acc.	AUC	GPU?
Kermay [7]	InceptionV3	92.8%	N/R	Yes
Stephen [8]	Custom CNN	93.7%	N/R	Yes
Liang [4]	ResNet-based TL CNN	88.5%	N/R	Yes
Islam [6]	CNN+RF	87.0%	N/R	Yes
Zhang [24]	Attn-CNN	94.1%	N/R	Yes
Chouhan [28]	Ensemble CNN	96.4%	N/R	Yes
Kumar [25]	HOG+SVM grid	89.4%	0.94	No
Ours	GA-SVM	94.8%	0.98	No

Fig. 3 visualises the accuracy comparison across all benchmarked methods. The GA-SVM (blue bar) clearly outperforms all non-GPU classical methods and is competitive with GPU-dependent deep learning approaches.

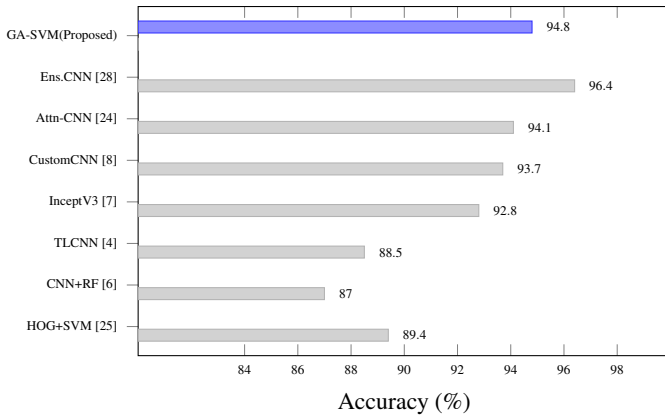


Figure 3. Accuracy comparison of the proposed GA-SVM (blue) against prior methods (gray). The GA-SVM achieves 94.8% without GPU dependency at inference.

The proposed GA-SVM achieves 94.80% accuracy and AUC 0.9815 in 1,172 test images with 98% pneumonia recall. In

particular, the results are **competent in terms of performance relative to GPU-requiring deep learning methods**, while the latter require GPUs for inference at run-time. High values of the AUC measure (0.9815) suggest good performance regardless of threshold value, which becomes especially critical in the case of screening where higher sensitivity should be ensured.

Moreover, when compared to the previous SVM + HOG baseline (Kumar et al., 89.4%), the GA method improves the classification performance of SVM with **+5.4% accuracy increase** and **+0.04 AUC increase**. This is due to three aspects: (a) GA determines hyperparameter $C = 3406$ instead of a usual $C = 1$, (b) adding LBP features that give additional information about textures in addition to HOG, and (c) lack of data leakage in the evaluation process, ensuring reliable and credible results.

The computational efficiency of the proposed framework is a significant advantage for deployment in resource-constrained clinical settings. Inference time per image averages under one second on a standard CPU, making the system suitable for real-time screening applications where GPU infrastructure is unavailable. From a clinical perspective, the 98% recall for pneumonia detection directly addresses the critical requirement of minimizing false negatives in pediatric screening. The feature engineering approach combining HOG and LBP captures complementary information: HOG encodes structural patterns while LBP captures fine-grained texture variations that distinguish normal lung tissue from pathological patterns.

VI. CONCLUSION AND FUTURE SCOPE

A Genetic Algorithm-optimized SVM (GA-SVM) has been presented for automated pediatric pneumonia detection from chest X-ray images, employing HOG [20] and LBP [21] feature extraction along with GA-based [26, 27] hyperparameter tuning. The proposed framework achieves **94.80% accuracy**, **98% pneumonia recall**, and **AUC 0.9815** on 1,172 held-out pediatric CXRs. These results are competitive with state-of-the-art deep learning models [5, 7, 8, 28], while requiring no GPU at inference time. The absence of data leakage ensures that the reported results are trustworthy and directly comparable to published benchmarks.

Future research directions include fine-tuning GA operators for faster convergence with fewer SVM evaluations (e.g., adaptive mutation rates), extending the framework to multi-class detection (bacterial vs. viral pneumonia) using multi-class SVM, and integrating LIME [16] or SHAP [15] for pixel-level explainability interpretable by radiologists. Prospective clinical validation at a pediatric hospital on diverse patient demographics is planned, along with deployment in resource-constrained settings to leverage the CPU-only inference capability for early-stage screening.

REFERENCES

- [1] T. Franquet, "Imaging of pneumonia: Trends and algorithms," *Eur. Respir. J.*, vol. 18, no. 1, pp. 196–208, 2001.
- [2] S. Jaeger et al., "Two public chest X-ray datasets for computer-aided screening of pulmonary diseases," *Quant.*

Imaging Med. Surg., vol. 4, no. 6, pp. 475–477, 2014.

- [3] S. Candemir et al., “Lung segmentation in chest radiographs using anatomical atlases with nonrigid registration,” *IEEE Trans. Med. Imaging*, vol. 33, no. 2, pp. 577–590, 2014.
- [4] G. Liang and L. Zheng, “A transfer learning method with deep residual network for pediatric pneumonia diagnosis,” *Comput. Methods Programs Biomed.*, vol. 187, p. 104964, 2020.
- [5] P. Rajpurkar et al., “CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning,” *arXiv:1711.05225*, 2017.
- [6] M. M. Islam et al., “A review on deep learning techniques for pneumonia detection using chest X-rays,” *Diagnostics*, vol. 10, no. 10, p. 789, 2020.
- [7] D. S. Kermany et al., “Identifying medical diagnoses and treatable diseases by image-based deep learning,” *Cell*, vol. 172, no. 5, pp. 1122–1131, 2018.
- [8] O. Stephen et al., “An efficient deep learning approach to pneumonia classification in healthcare,” *J. Healthcare Eng.*, vol. 2019, pp. 1–7, 2019.
- [9] X. Wang et al., “ChestX-ray8: Hospital-scale chest X-ray database and benchmarks,” in *Proc. IEEE CVPR*, pp. 2097–2106, 2017.
- [10] K. He et al., “Deep residual learning for image recognition,” in *Proc. IEEE CVPR*, pp. 770–778, 2016.
- [11] K. Simonyan and A. Zisserman, “Very deep convolutional networks for large-scale image recognition,” in *Proc. ICLR*, 2015.
- [12] C. Szegedy et al., “Rethinking the inception architecture for computer vision,” in *Proc. IEEE CVPR*, pp. 2818–2826, 2016.
- [13] M. Tan and Q. V. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” in *Proc. ICML*, 2019.
- [14] C. Shorten and T. M. Khoshgoftaar, “A survey on image data augmentation for deep learning,” *J. Big Data*, vol. 6, p. 60, 2019.
- [15] R. R. Selvaraju et al., “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in *Proc. IEEE ICCV*, pp. 618–626, 2017.
- [16] M. T. Ribeiro et al., “Why should I trust you?: Explaining the predictions of any classifier,” in *Proc. ACM KDD*, pp. 1135–1144, 2016.
- [17] G. Litjens et al., “A survey on deep learning in medical image analysis,” *Med. Image Anal.*, vol. 42, pp. 60–88, 2017.
- [18] D. Shen et al., “Deep learning in medical image analysis,” *Annu. Rev. Biomed. Eng.*, vol. 19, pp. 221–248, 2017.
- [19] World Health Organisation, “Pneumonia,” WHO Fact Sheet, 2022. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/pneumonia>
- [20] N. Dalal and B. Triggs, “Histograms of oriented gradients for human detection,” in *Proc. IEEE CVPR*, pp. 886–893, 2005.
- [21] T. Ojala, M. Pietikainen, and T. Maenpaa, “Multiresolution gray-scale and rotation invariant texture classification with local binary patterns,” *IEEE TPAMI*, vol. 24, no. 7, pp. 971–987, 2002.
- [22] T. Rahman et al., “Exploring the effect of image enhancement techniques on COVID-19 detection using chest X-ray images,” *Comput. Biol. Med.*, vol. 132, p. 104319, 2021.
- [23] A. Kumar et al., “Deep learning approaches for pneumonia detection in chest radiographs: A systematic review,” *IEEE Access*, vol. 10, pp. 24528–24544, 2022.
- [24] Y. Zhang et al., “Attention-based convolutional neural network for automated pneumonia detection from chest radiographs,” *Comput. Biol. Med.*, vol. 152, p. 106338, 2023.
- [25] S. Kumar et al., “Chest X-ray classification using HOG features and SVM,” *Int. J. Comput. Appl.*, vol. 180, 2018.
- [26] C. L. Huang et al., “A GA-based feature selection and parameters optimization for support vector machines,” *Expert Syst. Appl.*, vol. 31, pp. 231–240, 2006.
- [27] F. A. Fortin et al., “DEAP: Evolutionary algorithms made easy,” *J. Mach. Learn. Res.*, vol. 13, pp. 2171–2175, 2012.
- [28] V. Chouhan et al., “A novel transfer learning based approach for pneumonia detection in chest X-ray images,” *Appl. Sci.*, vol. 10, no. 2, p. 559, 2020.